

Dr Thomas Frost

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PROFILE

Oxford-trained physician and machine learning researcher, concluding a PhD in AI-enabled healthcare at UCL. I specialise in reinforcement learning for high-stakes clinical systems, with my research spotlighted on the [TalkRL podcast](#). During my PhD, I also released [Insulin4RL](#), an open-source dataset for healthcare-based reinforcement learning; supported the development of [FlowEHR](#), a local hospital platform for ML development and deployment; and provided formal evaluations of LLM discharge summaries.

EDUCATION

University College London (UCL) London, UK
PhD in AI-enabled Healthcare Sept 2022 — Expected Oct 2026

- **Thesis:** “Offline reinforcement learning for clinical data with natural timings.”
- Designed a novel machine learning system for ICU infusion dose recommendations using complex clinical data.
- Created and published [Insulin4RL](#), a freely available dataset for reinforcement learning using naturally timed health data.
- Partnered with Microsoft and UCLH to help refine [FlowEHR](#), a MLOps project for deployment of clinical ML models.

University College London (UCL) London, UK
MRes (Master of Research) in AI-enabled Healthcare Sept 2021 — Sept 2022

University of Oxford Oxford, UK
BM BCh (Bachelor of Medicine, Bachelor of Surgery) Oct 2012 — Jul 2018

University of Oxford Oxford, UK
MA in Medical Sciences, First Class Oct 2014 — Jul 2015

WORK EXPERIENCE

Emergency Medicine Clinician London, UK
Homerton University Hospital Sept 2021 — Present

- Delivering frontline care in high-pressure environments, routinely making safety-critical decisions with incomplete data.
- Coordination of multidisciplinary teams for management of trauma and resuscitation for unwell patients.

Junior Clinical Fellow in Emergency Medicine Edinburgh, UK
NHS Lothian Aug 2020 — Jul 2021

Foundation Doctor Edinburgh, UK
NHS Lothian Aug 2018 — Jul 2020

CERTIFICATES & QUALIFICATIONS

Member of the Royal College of Emergency Medicine Jan 2024
Royal College of Emergency Medicine

TECHNICAL SKILLS

- **Core:** Python, PyTorch, and data science suite (NumPy/Pandas/Polars/SciPy).
- **Advanced ML:** Reinforcement Learning, Time-Series Forecasting, Clinical Decision Support.
- **Cloud & Infrastructure:** AWS Compute (EC2/S3 for research workloads), FlowEHR (TRE User/Validator).
- **Clinical Products:** Human-in-the-loop Evaluation, Clinical Safety Metrics, Medical Data Privacy.

AWARDS AND RECOGNITION

- **TalkRL Podcast Guest:** Interviewed on [TalkRL](#), which has featured guests from DeepMind, OpenAI, and Meta.
- **UCL Three-Minute Thesis:** University finalist, communicating complex AI concepts to non-technical audiences.
- **UCL Centre of Digital Innovation Scholar:** Awarded £5,000 AWS research credits + cloud computing mentorship.
- **UKRI Studentship:** Full PhD stipend + annual research budget for outstanding research potential.
- **Lord Crewe Scholar (Oxford):** Awarded for academic distinction.

SELECTED PUBLICATIONS

- Frost, T.** & Harris, S. (2026). Insulin4RL: Real-time insulin management in the intensive care unit for offline reinforcement learning. Preprint arXiv:2606.19481.
- Frost, T.** & Harris, S. (2026). Insulin4RL: Real-time insulin infusions for offline reinforcement learning. PhysioNet.
- Frost, T.**, Vaidya, H. & Harris, S. (2026). The hidden risks of temporal resampling in clinical reinforcement learning. Preprint arXiv:2602.06603.
- Frost, T.**, Li, K. & Harris, S. (2025). Robust real-time mortality prediction in the intensive care unit using temporal difference learning. In: Proceedings of the 4th Machine Learning for Health Symposium, PMLR, pp. 350–363.
- Ellershaw, S., Tomlinson, C., Burton, O.E., et al. (2024). Automated generation of hospital discharge summaries using clinical guidelines and large language models. In: AAAI 2024 Spring Symposium on Clinical Foundation Models.
- Collins, D.R.J., Jobanputra, K., **Frost, T.**, et al. (2017). Cardiovascular disease risk and prevention amongst Syrian refugees: Mixed methods study of Médecins Sans Frontières programme in Jordan. *Conflict and Health*, 11(1), 14.
- Frost, T.**, Sinha, D. & Gilbert, B.J. (2014). Should assisted dying be legalised?. *Philosophy, Ethics, and Humanities in Medicine*, 9(1), 3.